# Morphology-Aware Modeling of Code-Mixed Indian Languages: A Bengali-Hindi Case Study

B.Tech Project Report

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**Abstract**

This work presents a morphology-aware text analyzer for Bengali-Hindi code-switched speech, targeting a specific and under-studied population: Bengali L1 speakers who use Hindi as L2 in institutional settings. We collect and annotate a corpus of 42.8 minutes of naturalistic speech from 8 campus staff members at IIT Kharagpur (2,060 analysed participant words), documenting systematic L1 transfer patterns including masculine-default gender errors, near-complete avoidance of the ergative marker *ne*, and multi-level code-switching from tag-switches to full inter-sentential Bengali clauses. Our rule-based analyzer operates on Roman-transliterated text and detects gender agreement violations, ergative gaps, and code-switch boundaries with reference to curated Hindi–Bengali lexicons. We describe an iterative error audit in which every one of the analyzer's initial 36 gender-error detections was checked against its transcript context, 17 false positives were traced to six distinct causes, and the rules were revised until the final 21 detections were all confirmed. A key finding is that ASR normalization of non-standard forms is **architectural, not a matter of model size or finetuning**: a four-way comparison of human-corrected transcripts, Whisper-large-v3, IndicWhisper-small and IndicWhisper-large shows that the ASR systems hide 90–100% of the gender errors, 38–90% of the code-switches and 67–94% of the Bengali words. Even the large Hindi-finetuned model (1.5B parameters) preserves only 2 of 21 errors, while introducing hallucination loops on code-switched audio. Morphological analysis of L1 transfer in code-switched speech therefore requires human-corrected transcripts.

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# 1 Introduction

Code-switching—the alternation between two or more languages within a single conversation or utterance—is a pervasive feature of multilingual communities worldwide [Poplack, 1980, Myers-Scotton, 1997]. In India, where speakers routinely navigate between regional languages, Hindi, and English, code-switching is not a sign of linguistic deficiency but a communicative resource [Bhatia and Ritchie, 2004].

A particularly interesting case arises when speakers of a language without grammatical gender (such as Bengali) use a language that requires it (such as Hindi). Bengali marks neither gender on nouns nor gender agreement on verbs, while Hindi has a two-gender system (masculine/feminine) with obligatory agreement between nouns and verbs across tense, aspect, and number [Kachru, 2006]. When Bengali L1 speakers produce Hindi L2, they must map a genderless system onto a gendered one—a task that produces systematic and predictable errors.

This B.Tech project addresses three interconnected questions:

1. What are the systematic morphological patterns of L1 (Bengali) transfer in L2 (Hindi) speech among Bengali-dominant speakers in an institutional setting?
2. Can a rule-based text analyzer reliably detect these patterns in transcribed speech, and how can its precision be measured and improved?
3. How does the choice of ASR frontend (standard Whisper, Hindi-finetuned Whisper, or human correction) affect the visibility of these patterns to downstream analysis?

## 1.1 Motivation

Most NLP work on code-switching in Indian languages focuses on social media text [Bali et al., 2014] or targets Hindi-English mixing [Singh et al., 2018]. Bengali-Hindi code-switching in *spoken* language is comparatively under-studied, despite being common in eastern India where Bengali speakers interact with Hindi-dominant institutions. The IIT Kharagpur campus, located in West Bengal with a Hindi-medium administrative culture and Bengali-speaking staff, provides a natural laboratory for this study.

A second motivation is methodological. Off-the-shelf ASR systems like Whisper are trained on standard language and silently normalize non-standard forms. For sociolinguistic and L1-transfer research, this normalization destroys the very data of interest. We quantify this destruction and argue for human-in-the-loop transcription pipelines.

A third motivation is practical. The campus staff who participated in this study use Hindi daily with students, colleagues and visitors. Documenting the regularities in their speech, rather than treating them as random mistakes, is a precondition for any tool—an ASR system, a learning aid, an accessibility service—that is meant to work for them.

## 1.2 Hypotheses

We organise the analysis around four testable hypotheses, which are revisited in Section 11:

**H1 (Transfer)**
Because Bengali lacks grammatical gender, gender-agreement errors in the speakers' Hindi will be predominantly in the direction of the unmarked masculine form.

**H2 (Avoidance)**
Because Bengali lacks ergative case marking, speakers will avoid or under-produce the ergative marker *ne* in transitive perfective contexts.

**H3 (Salience)**
Bengali lexical insertions will be concentrated in the domain of household objects, where the

first language is the language of daily experience, and will decrease in the metalinguistic and conversational sections.

**H4 (Opacity)**
Standard and Hindi-finetuned ASR will hide the above phenomena irrespective of model size, because they are trained to produce standard-language output.

### 1.3 Contributions

- A curated corpus of 8 speakers (4 female, 4 male), 42.8 minutes of audio across 9 recordings, 2,060 analysed participant words (2,990 tokens including the data collector's prompts), with human-corrected Roman transliterations preserving all non-standard forms.
- A catalog of 6 morphological transfer rules (R1–R6) grounded in the data, covering gender agreement, ergative case, and code-switching.
- A rule-based `MorphologyAwareAnalyzer` implemented in Python, with curated Hindi–Bengali lexicons, that detects gender errors, ergative avoidance, and classifies code-switches.
- A documented iterative error audit (36 initial detections → 17 false positives characterised → 21 verified detections) that turns the analyzer's output into a measured precision/recall estimate rather than an unexamined count.
- A four-way ASR comparison (human-corrected, Whisper-large-v3, IndicWhisper-small, IndicWhisper-large) showing that ASR systems hide 90–100% of gender errors regardless of model size or Hindi finetuning, and that the large Hindi model preserves only 2/21 errors while introducing hallucination on code-switched segments.

### 1.4 Scope and Organisation of this Report

The study is a descriptive case study, not a population survey: with eight speakers and roughly two thousand analysed words, it documents patterns and tests a measurement approach rather than estimating population parameters. Section 2 sets out the linguistic background; Section 3 reviews related work; Section 4 describes the corpus and the transcription protocol; Section 5 presents a profile of each speaker; Section 6 lists the transfer rules; Section 7 describes the analyzer; Section 8 reports the error audit; Section 9 and Section 10 present the evaluation and the ASR comparison; Section 11 discusses the findings and their limits; and Section 12 concludes. Appendices give the full list of detections, the ergative contexts, the audit table, lexicon summaries and reproduction instructions.

## 2 Linguistic Background

### 2.1 Gender and Agreement in Hindi

Hindi nouns are either masculine or feminine, and the gender of a noun is reflected on adjectives, postpositions such as *ka/ki/ke*, and—most importantly for this study—on the verb. In the perfective, the participle ends in *-a* (masculine) or *-i* (feminine): *kursi toot gayi* "the chair broke" (F) versus *darwaza toot gaya* "the door broke" (M). In the habitual and progressive aspects, the verb agrees with the subject: *woh padhta hai* (M) versus *woh padhti hai* (F). Speakers whose first language has no gender system must learn both the gender of each noun, which is largely arbitrary for inanimates, and the agreement rules that propagate it.

Two points matter for annotation. First, the gender of a noun is a property of the lexicon: *chawal* "rice" and *glass* are masculine, while *roti* "bread" and *kitaab* "book" are feminine. An analyzer that gets the gender of a noun wrong will produce false errors (Section 8 documents exactly this). Second, masculine agreement is the standard form for mixed-gender groups and for impersonal constructions, so a masculine verb is not an error when the referent is a group of boys

and girls or when no specific person is referred to.

### 2.2 Gender in Bengali

Bengali has no grammatical gender. Nouns do not carry a gender class, adjectives do not agree with nouns, and verbs inflect for person, tense, aspect and honorific level but not for the gender of the subject. The Bengali sentence *boiti ulto kore rekhechi* "I have kept the book upside down" is identical whether the speaker is male or female. A Bengali L1 speaker producing Hindi therefore has no gender category to transfer and must acquire one; the typical outcome in the second-language literature is a default to a single unmarked form [Sabourin et al., 2006, Grüter et al., 2012].

Bengali does have grammatical features that surface in the code-switched data. The classifiers *-ta*, *-ti*, *-khana* follow numerals and nouns (*ek ta chair*), the copula *ache* corresponds to Hindi *hai*, the present-progressive endings *-che/-chi/-cche/-cchi* correspond to Hindi *raha/rahi hai* (*jol dhalche* "water is being poured"), and the perfective endings *-lam/-lo/-le* mark first-, third- and second-person past. These are the surface markers the analyzer uses to recognise Bengali stretches (Section 7).

### 2.3 Ergativity

In Hindi the perfective of transitive verbs is ergative: the agent is marked with the postposition *ne* and the verb agrees with the object rather than the subject, as in *maine kitaab padhi* "I read the book" (verb feminine, agreeing with *kitaab*). Ergative alignment is a well-known difficulty for second-language learners [Butt, 1995]. Bengali has no ergative marking: the agent is in the nominative, so *ami boi porlam* has the same structure whether the verb is perfective or not. The *ne* construction is thus a candidate for avoidance, in the sense of Schachter [1974]: learners may sidestep a structure they find difficult rather than produce it incorrectly.

### 2.4 Code-Switching Frameworks

Poplack [1980] distinguishes tag-switching, intra-sentential switching and inter-sentential switching, a typology we adopt for classification. Muysken [2000] proposes a complementary typology of insertion, alternation and congruent lexicalisation, which we use in the discussion to ask whether the Bengali material in our data behaves like insertion of single items into a Hindi frame or like alternation between two clauses. Myers-Scotton [1997]'s Matrix Language Frame model adds the notion of a matrix language that supplies the grammatical frame; in our data Hindi is the matrix language of the interview, with Bengali supplying content words and, in some speakers, full clauses.

### 2.5 Transliteration and Glossing Conventions

All transcripts used for analysis are in Roman transliteration, following informal conventions that match how Hindi is typed on phones: long vowels are not marked, aspirated consonants are written with *h*, and nasalisation is mostly omitted. Variant spellings are normalised only where the lexicon requires (for example *kitaab/kitab*, *khursi/kursi*); we do *not* normalise grammatical forms. Devanagari output from the ASR systems is retained alongside the Roman form and is converted to Roman only for lexicon matching (Section 7). In examples, M/F mark masculine/feminine agreement and CLF a classifier; uncertain or inaudible stretches are written `[unclear]`.

# 3 Related Work

## 3.1 Code-Switching in Indian Languages

Poplack [1980] established the foundational typology of code-switching: tag-switching (insertion of a discourse marker or tag from one language), intra-sentential switching (switching within a clause), and inter-sentential switching (switching at clause or sentence boundaries). This typology remains the standard framework and is adopted in our analysis.

Research on Hindi-English code-switching is extensive, driven by social media data [Bali et al., 2014, Singh et al., 2018, Khanuja et al., 2020]. The GLUECoS benchmark [Khanuja et al., 2020] provides evaluation tasks for code-mixed NLP but focuses on Hindi-English text data rather than spoken language with morphological transfer.

Bengali-Hindi code-switching has received less attention. Muysken [2000] discusses insertion patterns in South Asian language pairs, and some work exists on Bengali-English mixing [Das and Gambäck, 2014], but the specific case of Bengali L1 transfer in spoken Hindi L2 is largely unaddressed in the computational linguistics literature.

## 3.2 L1 Transfer and Gender Agreement

The acquisition of grammatical gender by speakers of genderless L1s is well-studied in European language pairs [Sabourin et al., 2006, Grüter et al., 2012]. The consistent finding is that speakers default to the unmarked form—typically masculine in languages like Hindi, French, and Spanish. Our data are consistent with this pattern: all 21 gender errors detected in the corpus are masculine-default, with a single candidate instance of the reverse direction found by manual inspection (Section 11).

For Hindi specifically, Sharma [2004] documents gender errors in Hindi L2 speakers, and the ergative construction has been studied as a known difficulty for L2 learners [Butt, 1995]. Our contribution is the specific documentation of Bengali-to-Hindi transfer patterns with corpus evidence.

## 3.3 ASR for Code-Switched Speech

Whisper [Radford et al., 2023] and its multilingual variants handle many languages but are trained primarily on monolingual data. For code-switched speech, specialized systems like those from the MUCS shared task [Diwan et al., 2021] and IndicWhisper [Jain et al., 2024] attempt to handle language mixing. However, even these systems are not designed to preserve morphological errors—they aim for "correct" transcription, which in our case means hiding the phenomena under study.

# 4 Corpus and Data Collection

## 4.1 Study Design

Data was collected from 8 campus staff members at IIT Kharagpur, all Bengali L1 speakers who use Hindi as L2 in their work environment. The study was designed as a structured elicitation task with six sections:

Table 1: Questionnaire sections and target phenomena

| Section | Task | Target | Elicits |
|---|---|---|---|
| A | Object naming | Vocabulary | Bengali lexical items |
| B | Photo comparison | Adjectives | Adjective agreement |
| C | Action description | Verbs | Object-verb gender agreement |
| D | State description | Intransitives | Subject-verb agreement |
| E | Free conversation | Spontaneous | Natural code-switching |
| F | Background | Metalinguistic | Language attitudes |

Sections A–D use a set of picture stimuli. In the naming task the participant is shown household objects—a door, chair, fan, book, house, window, glass, plate of food, car, key, hand—and asked to say what each one is. In the comparison and description tasks the participant is shown pairs of photographs and asked what is different ("*in dono mein phark kya hai?*") and what the person in the photograph is doing ("*iss photo mein kya kar raha hai?*"). The stimuli were chosen because they elicit the two agreement environments of interest: feminine nouns that take a verb (*kitaab padh rahi hai*, *khursi toot gayi*) and masculine nouns that serve as controls. Sections E and F are conversational: participants talk about their work, family, schooling and the languages they use at home and on campus.

A note on the elicitation. The data collector asks the questions in Hindi and addresses the participant respectfully ("*dada*", "*didi*", "*aap*"); the participants answer in Hindi, switching into Bengali where they choose. The collector's turns are marked in the transcripts (`c:`) and are excluded from every count in this report; only participant turns (`s:`) are analysed.

### 4.2 Participants and Recordings

Eight participants (S1–S8) contributed nine recordings; S1 recorded twice. Table 2 summarises the recordings together with their acoustic quality measured from the original media.

Table 2: Recordings and acoustic quality (from the technical inventory)

| Spk. | File | Dur. (s) | SNR (dB) | Speech frac. | Verdict |
|---|---|---|---|---|---|
| S1 | IMG_6647 | 231.2 | 14.1 | 0.34 | Clean |
| S1 | IMG_6648 | 76.7 | 13.1 | 0.40 | Clean |
| S2 | IMG_6650 | 309.4 | 19.3 | 0.49 | Clean |
| S3 | IMG_6651 | 370.9 | 19.4 | 0.50 | Clean |
| S4 | IMG_6652 | 341.0 | 8.4 | 0.22 | Usable, noisy |
| S5 | IMG_6654 | 387.7 | 13.5 | 0.39 | Clean |
| S6 | IMG_6660 | 304.7 | 12.8 | 0.33 | Clean |
| S7 | NewRecording7 (audio) | 330.1 | 17.7 | 0.47 | Clean |
| S8 | IMG_6664 | 216.4 | — | — | Not in inventory |
| **Total** | | **2,568.1** | | | 42.8 min |

The interviews in the technical inventory were recorded as 4K video (HEVC, AAC audio at 48 kHz) with the exception of one participant (S7), who declined to be filmed and was recorded audio-only, which we treat as an expression of the voluntary nature of participation. No file shows clipping. Four short clips (3–30 seconds: camera checks and false starts) were discarded as too short for analysis. The speech fraction of 22–50% reflects the interview format, with long pauses and collector turns. S4's recording has the lowest signal-to-noise ratio (8.4 dB, background noise) and is the one most likely to contain mis-transcribed material; we therefore examine S4's detections with particular care in Section 8.

### 4.3 Privacy and Ethics

Participants are identified in all output only by speaker codes (S1–S8). Where a real name is spoken in a recording it is replaced by `[NAME]` in the transcript. The audio and video files are kept in a private repository whose ignore rules exclude the recordings and any participant-identifying material from version control. Participants are campus staff in an employment relationship with the institution, and the elicitation tasks were designed to be neutral and non-evaluative; nevertheless, the topic—errors in a language used at work—is potentially sensitive, which is why the individual profiles in Section 5 describe patterns in the speech rather than evaluating anyone's competence.

### 4.4 Transcription Pipeline

1. **Audio extraction**: WAV files (16 kHz, mono) extracted from the interview recordings with `ffmpeg`.
2. **Initial ASR**: `faster-whisper` with the Whisper large-v3 model (int8 quantisation for CPU), language forced to Hindi, word-level timestamps, voice-activity filtering and beam search (width 5). The prompt instructs the model to preserve hesitations and false starts and not to normalise dialect forms.
3. **Human correction**: Each segment is manually corrected against the audio, preserving all non-standard forms, hesitations, and code-switches. Bengali words are transliterated to Roman script with glosses.
4. **Annotation**: Gender errors, code-switches, and Bengali insertions are marked in annotator notes and then analysed by the rule-based system.

### 4.5 Transcription Conventions

The corrected transcripts follow a small set of principles, applied throughout the corpus:

- The Whisper output is a draft, never a final transcript. Every machine transcript carries an explicit "unverified" warning until a human has checked each segment against the audio.
- Nothing is cleaned up or normalised. False starts, repetitions, hesitations (*uhhh*, *kya bole*) and apparent errors are kept exactly as produced, because they are the data.
- Audio that cannot be made out is not filled in; it is written `[unclear]` with its timestamp.
- Where it is unclear whether a word is Bengali or Hindi, it is flagged rather than decided; the language-identification step reports such words as *unknown* rather than guessing.
- Speakers are identified by codes, and spoken names are replaced by `[NAME]`.
- Devanagari output is retained, with the Roman transliteration alongside it.

### 4.6 Human Verification and Whisper Errors

A critical finding during transcription: Whisper systematically "corrects" the data:

- Gender errors are silently fixed (e.g., speaker says *tut gaya*, Whisper outputs the "correct" *tut gayi*).
- Bengali words are converted to similar-sounding Hindi (e.g., *dorja* → *darwaza*).
- Code-switched passages are rendered as Hindi approximations.

Of the 183 segments across all recordings, 171 were verified by a human listener: 144 contain participant speech, 22 contain only the collector's speech, and 5 are noise. Of the 144 participant-speech segments, 120 required correction (83.3%) and 24 were verified correct as transcribed. Word error rate is not computable because Whisper outputs Devanagari while the corrections are in Roman, so segment-level error rate is used instead. The error types are dominated by wrong

words (116 of 120 corrected segments, 96.7%), followed by missed Bengali (2), hallucination (1) and other (1).

### 4.7 Corpus Statistics

Table 3: Corpus statistics per speaker (corrected transcripts; participant turns only)

| Spk. | Gender | Role | Dur. (min) | Segs | Words |
|---|---|---|---|---|---|
| S1 | M | Cycle shop | 5.1 | 14 | 178 |
| S2 | F | Staff | 5.2 | 12 | 202 |
| S3 | F | Staff | 6.2 | 15 | 230 |
| S4 | F | Staff | 5.7 | 15 | 307 |
| S5 | M | Mess staff | 6.5 | 17 | 307 |
| S6 | M | Mess staff | 5.1 | 15 | 299 |
| S7 | M | Security | 5.5 | 19 | 308 |
| S8 | F | Staff | 3.6 | 13 | 229 |
| **Total** | | | **42.8** | **120** | **2,060** |

The analysed sample (2,060 words) is smaller than the raw token count of the corrected transcripts (2,990 tokens) because the collector's questions and annotation markers are excluded, and smaller than the raw Whisper word count (2,957) because raw ASR text does not separate the collector's turns from the participant's, so its word count also includes the collector's speech. The number of words per speaker ranges from 178 to 308, which sets the scale at which per-speaker comparisons should be read.

## 5 Speaker Profiles and Case Studies

Table 4: Speaker demographics and error counts (21 detected gender errors; 159 Bengali-word candidates)

| Speaker | Gender | Role | Words | Gender Errors | Per 1,000 words | Bengali Words |
|---|---|---|---|---|---|---|
| S1 | M | Cycle shop | 178* | 1 | 5.6 | 15 |
| S2 | F | Staff | 202 | 3 | 14.9 | 34 |
| S3 | F | Staff | 230 | 4 | 17.4 | 2 |
| S4 | F | Staff | 307 | 11 | 35.8 | 9 |
| S5 | M | Mess staff | 307 | 0 | 0.0 | 23 |
| S6 | M | Mess staff | 299 | 0 | 0.0 | 53 |
| S7 | M | Security | 308 | 1 | 3.2 | 16 |
| S8 | F | Staff | 229 | 1 | 4.4 | 7 |
| **Total** | | | **2,060** | **21** | **10.2** | **159** |

*S1 has two recordings (6647 + 6648); word count is the combined total.

Key observations from speaker profiles:

- Gender errors are concentrated in female speakers (S2, S3, S4, S8), who account for 19 of the 21 detections (19.6 per 1,000 words against 1.8 for male speakers). Both male-speaker detections (S1, S7) are noun-verb errors in the photo-description task; the subject-verb rule (R2) applies only to female speakers, for whom a masculine self-reference is non-standard.

- S4 shows the highest error rate (35.8 per 1,000 words), consistent with later Hindi acquisition.
- S6 uses the most Bengali words (53) and produces full Bengali clauses, representing the high-Bengali end of the proficiency gradient.
- Male speakers (S1, S5, S7) show few gender errors because masculine default happens to be correct for male speakers' self-reference.

The remainder of this section describes each speaker in turn, using excerpts from the corrected transcripts (Roman transliteration; the English gloss is ours). Because the detector has known limits (Section 8), the descriptions rely on the verified detections and on direct reading of the transcripts rather than on counts alone.

### 5.1 S1 — Male, cycle shop

S1 has two recordings, 5.1 minutes in total. In the object-naming and description tasks S1 mixes Hindi, Bengali and English within one list, describing the pictures as:

> *eh bhat kha raha hai, dorja khol raha hai, jol dhalche, jol gir gaya, eh copy khula hai, chair bhenge ache, fan bhanga*
> "this one is eating rice, opening the door, water is being poured, water fell, this notebook is open, the chair is broken, the fan is broken"

Here *bhat* "rice", *dorja* "door", *jol* "water", *dhalche* "is pouring" and *bhenge ache* "is broken" are Bengali. The one detected gender error is *roti kha raha hai* (feminine *roti* with a masculine progressive). S1 also describes his family and work: *mera cycle shop hai aar mera ek ladki ek ladka, dono study karta hai*, where the masculine singular verb with "a girl and a boy" is the standard default for a mixed pair and is therefore not counted. S1 says he studied B.Com for two years (*bcom hum 2 year padha tha*); both ergative-context verbs in his recordings (*padha*) appear without *ne*.

### 5.2 S2 — Female, staff

S2 has the second-highest share of Bengali words (34 candidate words, 16.8% of tokens) and produces all three inter-sentential switches outside S6. She shows visible lexical searching in Hindi (*eh dinner kya bole, uhhh kya bole*), which is preserved in the transcript. In the picture-description task she makes two noun-verb errors, *roti kha raha hai* and *kitaab padh raha hai*. In the conversational section she says of Telugu, *phir bhi thoda thoda word seekh raha hu*, a masculine progressive with a first-person auxiliary from a female speaker, which is a clear subject-verb error. S2's single self-referential error is a progressive; no habitual or perfective self-reference error was detected for her.

### 5.3 S3 — Female, staff

S3 uses the least Bengali (2 candidate words, 0.9%) and the most English (17.8%): her code-switching is mostly Hindi-English. She makes two picture-description errors (*kitaab padh raha hai*, *chair tut gaya*) and two self-referential errors while talking about her work and her Hindi: *jo bhi kaam aata hai hum log karta hai* "whatever work comes, we do" and *itna acchi bhi bol nahi pata hu* "I can't speak it that well", in which the habitual *pata* should be feminine *pati*. Importantly, S3 also produces correct feminine self-reference in the same recording (*mai khud banati hu* "I cook myself"), so her masculine forms are variable rather than categorical. One further form is a candidate counter-instance to the masculine-default pattern: *mera bada ladka school mein thi* (feminine *thi* with masculine *ladka*), which we discuss in Section 11.

### 5.4 S4 — Female, staff

S4 produces 11 of the 21 detected errors (52%). In the picture task she lists a series of states, *almari khul raha hai, kitaab padh raha hai, eh paani dal raha hai, glass tut gaya, kitaab ulta ho gaya, chair tut gaya, fan kharaab ho gaya*; of these, *almari khul raha*, *kitaab padh raha*, *kitaab ulta ho gaya* and *chair tut gaya* are errors, while *glass*, *paani* and *fan* are masculine and correctly agree. Her narrative about her work history is the richest source of subject-verb errors:

> *do hazaar pandrah mein kaam mein ghus gaya pehle se hum physiotherapy seekh raha ek accident mein mera kamar hi tut gaya isliye hum physiotherapy chodke idhar a gaya*
> "in 2015 I got into this work; I had been learning physiotherapy, in an accident my back broke, so I left physiotherapy and came here"

In this one sentence S4 produces three masculine self-referential verbs (*ghus gaya*, *seekh raha*, *a gaya*) and one noun-verb error (*kamar tut gaya*). She reports that her Hindi was learned at school (*hum jab 7th class aur 8th class padha aur do class mein hindi seekha*) and that she speaks mostly Bengali at home and with colleagues (*bengali mein hi baat karta hai, hindi thodi thodi hi use karta hai*). The recording has the lowest signal-to-noise ratio of the corpus, so its transcript was corrected with extra care.

### 5.5 S5 — Male, mess staff

S5 shows no gender errors, as expected for a male speaker's self-reference and with a correct noun agreement in the picture tasks. His speech is notable for dense mixing of Bengali function elements with Hindi and English: *yeh toh ek toh dorja rakhe ache* ("there is a door kept here") and, with a Bengali classifier and deictic around an English noun, *ek ta chair eta* ("this one chair"). Ten of his 14 detected switches are intra-sentential or three-way, i.e. Bengali elements inside Hindi/English frames rather than whole clauses. In the background section S5 reports having spent a long time in North India (*kaafi din mai bahar raha North India mein khas karke jaise ki Haryana Punjab Delhi*, "I was away for a long time in North India, especially Haryana, Punjab, Delhi") and says that he was obliged to speak Hindi there (*toh waha pe hindi bolna hi tha*); this broader Hindi exposure is a possible reason for his accuracy and is consistent with the proficiency gradient discussed in Section 11.

### 5.6 S6 — Male, mess staff

S6 produces the most Bengali material (53 candidate words, 17.7% of tokens) and the only sustained inter-sentential switches (5). In the naming task he lists objects in Bengali almost throughout: *dorja, kedara, pakha, boi, ghor, janala, jol bhora glass, parota, khabaar thaali.* In the action and state tasks he describes the pictures in full Bengali clauses, e.g. *boiti ulto kore rekhechi, chair bhenge geche, pakha bhenge geche.* S6 is at the high-Bengali end of the proficiency gradient; he is not an outlier but the extreme of a continuum along which the other speakers are arranged. In the background section he says that he learned Hindi while living at the institute, where he has been since childhood: *maine hindi iit mein rehke seekhi maane bachpan se iit mein hi hu.* The sentence is of particular interest because it contains the corpus' only attested ergative, *maine …seekhi*, with the verb correctly agreeing in the feminine with *hindi.*

### 5.7 S7 — Male, security

S7, the audio-only participant, has one noun-verb detection, *yeh ek ladies door khol raha hai*, which arises in a picture description where the feminine *ladies* is treated as the agent and takes a masculine progressive. S7 produces 7 three-way and 6 tag switches, mainly single Bengali nouns

inside English–Hindi lists (*house ka dorja*; *lunch ya dinner, bhat*), and almost no intra-sentential Bengali.

### 5.8 S8 — Female, staff

S8 has the shortest recording (3.6 minutes) and one detected error, in the self-reference *mai hi banata hai* "I'm the one who cooks", where the verb should be feminine *banati*. In the same recording she produces numerous correct feminine forms of self-reference: *jab tak mai idhar kaam pe rehti*, *Hindi use karti hu*, *mai subah ka duty pe aa jati hu*, *ek student ko padhati hu*. S8 is therefore a clear case of variable rather than categorical masculine default: four of her five first-person habitual forms are correct. In the naming task she names household objects in Hindi (*darwaza, fan, kitaab, ghar, khidki, glass, roti, khana, car, haath, chaabi*) with few Bengali items.

## 6 Morphological Rule Catalog

We identify six systematic transfer rules from the corpus. Counts below are the verified detections of the final analyzer (Section 8).

### 6.1 R1: Masculine Default on Feminine Nouns

Bengali has no grammatical gender. When producing Hindi, speakers default to the unmarked masculine verb form regardless of the noun's grammatical gender.

> **Produced:** *kitaab ulta ho gaya* (book upside-down became-M)
> **Expected:** *kitaab ulti ho gayi* (book upside-down became-F)

The corpus contains 12 detections of R1 from 5 speakers (S1, S2, S3, S4, S7), involving seven feminine nouns: *kitaab* (4), *roti* (2), *chair* (2), *almari* (1), *kamar* (1), *beti* (1) and *ladies* (1). They affect both the perfective (*tut gaya*, *ulta ho gaya*) and the progressive (*padh raha*, *kha raha*, *khol raha*, *khul raha*). Because R1 depends on the noun rather than the speaker, it appears in male as well as female speakers' picture descriptions. Of the 21 detections, all are masculine-default; manual inspection of the participant speech found one candidate reverse instance (Section 11). This asymmetry is diagnostic of L1 transfer from a genderless language, as opposed to random gender assignment.

### 6.2 R2: Masculine Default in Female Self-Reference

Female speakers use masculine verb forms when talking about themselves:

> **S4:** *kaam mein ghus gaya* (got-into-M work)
> **Expected:** *ghus gayi* (got-into-F)

There are 9 detections of R2 from 4 female speakers: S4 (5), S3 (2), S2 (1), S8 (1). They cover the perfective (*ghus gaya*, *a gaya*), the habitual (*karta*, *bolta*, *banata*, *pata hu*) and the progressive (*seekh raha*, *seekh raha hu*). S4's concentration (5 R2 and 6 R1 errors, 11 in total) is consistent with later Hindi acquisition; S2, despite being female, shows only one self-referential error (a progressive in the free-conversation section).

### 6.3 R3: Ergative *ne* Avoidance

Bengali has no ergative case marking. In Hindi, the ergative marker *ne* appears with transitive perfective verbs. The analyzer finds 8 candidate transitive perfective contexts across the corpus, of

which only one carries *ne* (S6, *maine hindi …seekhi*). We treat this as **near-complete avoidance**, but note that the number of contexts is very small and that some candidates are not clear transitive perfectives (for example S6 *janala khola ache*, where *khola* is the Bengali adjective "open", and S4 *bola* in a reported-speech frame); a manual audit of this rule remains to be done (Section 11.7).

This is an **avoidance strategy**, not an error: speakers restructure sentences to bypass the construction entirely, for example by using bare-subject forms such as *hum …padha tha* "(I) had studied" without an overt agent marker.

### 6.4 R4: Bengali Lexical Insertion (Tag-Switching)

Single Bengali content words inserted into Hindi matrix clauses, typically for household objects:

*yeh toh ek toh **dorja** rakhe ache* (this is a **door** kept here)

Common insertions: *dorja* (door), *janala* (window), *jol* (water), *bhat* (rice), *boi* (book), *pakha* (fan). The Bengali words are almost entirely nouns for household objects, which is the pattern predicted by H3.

### 6.5 R5: Bengali Inter-Sentential Switching

Full Bengali clauses between Hindi utterances, most frequent in S6:

**S6:** *cheleti porachuna korche* (the girl is studying)
**S6:** *boiti ulto kore rekhechi* (kept the book upside down)

S6 sits at the high-Bengali end of the proficiency gradient; S2 also produces inter-sentential switches, and S1 and S5 produce longer intra-sentential Bengali stretches.

### 6.6 R6: Three-Way Code-Mixing

Bengali, Hindi, and English elements combined within a single phrase:

**S5:** *ek **ta** chair **eta*** (one-CLF chair here)

Bengali classifier *ta* + English *chair* + Bengali deictic *eta*.

Table 5: Summary of the transfer rules and their counts in the corpus

| Rule | Phenomenon | Typical evidence | Count |
|---|---|---|---|
| R1 | Masculine default, noun-verb | *kitaab padh raha* | 12 |
| R2 | Masculine default, self-reference | *kaam mein ghus gaya* | 9 |
| R3 | Ergative *ne* avoidance | *hindi seekha* (no *ne*) | 7 of 8 contexts |
| R4 | Tag-/intra-sentential insertion | *dorja, jol, bhat* | 42 |
| R5 | Inter-sentential Bengali | *boiti ulto kore rekhechi* | 8 |
| R6 | Three-way mixing | *ek ta chair eta* | 28 |

## 7 System Architecture

### 7.1 Pipeline Overview

```
Audio (.wav)
```

```
  |
  v
ASR Frontend (Whisper / IndicWhisper)
  |
  v
[Human Correction] (corrected_text in Roman transliteration)
  |
  v
MorphologyAwareAnalyzer
  |-- Word-level Language ID
  |-- Gender Error Detection (R1, R2)
  |-- Ergative 'ne' Analysis (R3)
  |-- Code-Switch Classification (R4, R5, R6)
  |
  v
Linguistic Annotations (per-segment JSON)
```

The analyzer is a single Python class operating on one segment at a time. For each segment it first separates participant turns (`s:`) from collector turns (`c:`); only participant text is analysed, but the presence of a collector question addressed to the participant (a question containing *aap*) is recorded and used as evidence of self-reference (Section 7.4). Devanagari text is transliterated to Roman before analysis, so that the same lexicons serve both human-corrected and ASR text.

### 7.2 Lexicon Design

The analyzer relies on curated lexicons in Roman transliteration. Table 6 gives their sizes in the final version.

Table 6: Lexicon sizes (distinct lower-cased entries)

| **Lexicon** | **Entries** | **Examples** |
|---|---|---|
| Hindi feminine nouns | 81 | *kursi, chair, kitaab, roti, gaadi, khidki, almari* |
| Hindi masculine nouns | 36 | *darwaza, pankha, ghar, paani, khana, chawal* |
| Bengali words (with Hindi equivalent) | 62 | *dorja→darwaza, janala→khidki* |
| Bengali morphological markers | 15 | *-che, -chi, -cche, ache, ta, ti* |
| Masculine verb patterns | 68 | *gaya, ho gaya, karta, kha raha* |
| Feminine verb patterns | 68 | *gayi, ho gayi, karti, kha rahi* |
| Hindi function words | 252 | postpositions, pronouns, auxiliaries |
| English common words | 256 | *school, chair, duty, hospital* |

Three design decisions deserve comment. First, the noun lexicons are *hand-curated from the corpus* plus common household vocabulary, not drawn from a dictionary; this keeps the analyzer transparent but means that its recall is limited by what the lexicon knows. Second, English loanwords are assigned a Hindi gender by analogy with their Hindi equivalents (*chair* follows *kursi*, feminine; *glass* is masculine), which is a standard but not universal convention. Third, the lexicons were corrected as part of the audit: *chawal* and *glass* were wrongly listed as feminine and were removed from the feminine list, and *chair* was added (Section 8).

### 7.3 Word-Level Language Identification

A token is classified in a fixed order of precedence: (1) exact match in the Bengali word list; (2) Bengali morphological marker (*ache*, *ta*, …); (3) Bengali verb suffix (*-cchi, -che, -eche, -lam, -le,*

*-lo*) on a word of sufficient length; (4) English common word; (5) Hindi lexicon or verb pattern; (6) generic Hindi endings (*-ta, -ti, -ne, …*) on words longer than three letters; otherwise *unknown*. The *unknown* class is deliberately the default rather than a guess, in line with the rule that language identity should be flagged rather than decided when unclear. As Section 11.7 explains, the suffix heuristic in step (3) over-triggers on some Hindi words.

### 7.4 Gender Error Detection

**R1 (noun-verb).** For each occurrence of a feminine noun in the segment, the analyzer scans a five-word window for *compound* masculine verb forms (*tut gaya*, *ulta ho gaya*, *khol raha*, *padh raha*). Longest patterns are tried first, and each noun *occurrence* can raise at most one error. Only compound patterns are matched because single-word forms such as *karta* are too ambiguous: they may agree with an implicit subject rather than with the nearest noun.

**R2 (subject-verb).** Applied to female speakers only. A masculine verb form is flagged as a self-referential error only when *all* of the following hold:

1. the segment is not a photo description (a collector's *iss photo mein …* frame or the word *photo/picture*), where the verb describes the person in the picture;
2. the segment does not contain a mixed-gender group subject (*lerka lerki, ladka ladki*), for which masculine agreement is correct;
3. no third-person word (*woh, bachcha, pitaji, bhaiya, …*) occurs in the six words before the verb;
4. the word directly before a perfective *gaya/hua* verb is not an inanimate noun (*glass gir gaya* has *glass* as subject);
5. generic verbs (*hota, hota hai*) and impersonal contexts (*janam hua*) are excluded unless a first-person pronoun is immediately adjacent; and
6. there is positive evidence of self-reference: a first-person subject (*mai, main, hum, khud*) within six words before or after the verb, or, for habitual verbs, a collector question addressed to the participant in the same segment.

The ambiguous token *mein* is *not* treated as a first-person pronoun, because it is overwhelmingly the postposition "in".

**R2 (explicit first-person auxiliary).** A masculine participle directly followed by *hu/hun/hoon* (*seekh raha hu*, *bol nahi pata hu*) is unambiguous self-reference and is flagged for female speakers without further conditions.

**Deduplication.** If an R2 detection overlaps an R1 detection on the same verb (exact or substring match), only the more specific R1 is kept, and R2 is suppressed in listing-style segments with three or more R1 detections.

### 7.5 Ergative *ne* Analysis

A perfective form whose stem is in the transitive-verb list (*kha, pi, bol, sun, dekh, likh, padh, bana, seekh, khol, …*) is recorded as a candidate transitive perfective context. The analyzer looks five words backward for *ne* or one of the fused pronoun forms *maine, humne, tune, tumne, usne, isne, unhone, inhone, aapne*. The fused forms were added during the audit after a context with *maine* had initially been counted as lacking *ne*.

### 7.6 Code-Switch Classification

1. Words are classified as Hindi, Bengali, English, or unknown via lexicon lookup and morphological suffix matching.
2. Bengali word positions are grouped into contiguous runs (allowing gaps of up to 2 words for particles/determiners).
3. Runs are classified:
   - 1 word = **tag-switch** (R4)
   - 2–3 words = **intra-sentential** (R4)
   - 4+ words = **inter-sentential** (R5)
   - Bengali + English nearby = **three-way** (R6)

### 7.7 Devanagari Transliteration

Whisper outputs Devanagari script. For ASR comparison, a basic Devanagari-to-Roman transliterator maps characters to their informal Roman equivalents (e.g., क→*k*, ख→*kha*), with schwa deletion at word boundaries. This is approximate but sufficient for lexicon matching; it does mean that ASR text is analysed through an extra, lossy step (for example उलट may map to *ulat* rather than to the corrected *ulta*).

## 8 Iterative Error Audit

A rule-based analyzer's output is only as credible as its measured precision. An early version of the analyzer reported 36 gender errors, a number that was later set against an informally compiled list of 23 "manually counted" errors, and the discrepancy was unexplained. This section documents the audit that resolved it.

### 8.1 Method

Every one of the 36 detections was dumped with its rule, noun or subject, verb form, expected form and the full text of its segment, and each was classified as a true positive (a genuine masculine-default error by the speaker) or a false positive by reading the sentence in context and applying standard Hindi agreement. Where the segment alone was insufficient, the neighbouring segments and the collector's question were read. The audit was carried out by a single annotator; it therefore measures consistency with one reading of Hindi grammar, and a second annotator is the most important extension (Section 11.7).

## 8.2 Findings: Seventeen False Positives, Six Causes

Table 7: False-positive causes among the 36 initial detections

| Cause | N | Example and reason |
|---|---|---|
| Wrong noun gender in lexicon | 3 | *chawal kha raha hai*; *glass tut gaya*: *chawal* and *glass* are masculine in Hindi, so the verb is correct |
| Verb/noun homograph | 1 | *paani daal raha hai*: *daal* here is the verb "pour", not the noun "lentils" |
| Third-person subject | 4 | *pitaji gujar gaya*; *woh padhta hai* (about a male child); *woh kya bolta hai*: the subject is another person, not the speaker |
| Impersonal or generic construction | 2 | *Kharagpur mein janam hua* (birth happened; *janam* is masculine); *non-veg hota hai* (generic) |
| Mixed-gender group subject | 4 | *lerka lerki ek saath kaam karta tha / padhta tha*: masculine is the standard agreement for "boys and girls" |
| R1/R2 duplicate detection | 3 | the same verb counted twice, or flagged as R2 when already flagged as R1 |
| **Total false positives** | **17** | |

Of the 36 original detections, 19 were genuine, giving an initial precision of 53%. The causes are instructive: none is a failure of the idea of the rule but each is a failure of the knowledge the rule depends on—a noun's gender, whether a word is a verb or a noun, who the subject is, and whether the construction is personal at all. The full list is given in Appendix C.

## 8.3 Iterations

Table 8: Iterations of the analyzer and the number of gender errors detected

| Step | Change | Detections | Effect |
|---|---|---|---|
| 0 | Initial analyzer | 36 | 17 false positives |
| 1 | Remove *chawal, glass, daal* from the feminine lexicon; add a third-person subject check, a mixed-group exclusion and an impersonal-construction exclusion; deduplicate R1/R2 overlaps | 21 | removes lexicon and third-person errors |
| 2 | Skip inanimate subjects before perfective verbs; require a collector question addressed to the speaker for subjectless habitual forms | 19 | removes "glass gir gaya" and a stray R2 on a picture-description list |
| 3 | Stop treating *mein* as "I"; require first-person evidence within six words of the verb; restrict generic verbs | 16 | removes the last false positive (a sentence about a child's schooling); costs two true positives |
| 4 | Recall repairs: score each noun occurrence separately (a repeated *kitaab* was being skipped); add *chair* to the feminine lexicon; add the explicit first-person auxiliary rule | 21 | recovers *chair tut gaya* (S3, S4), *kitaab ulta ho gaya* (S4), *seekh raha hu* (S2), *pata hu* (S3) |

Step 3 shows a general trade-off. The most obvious way to remove a false positive about a third person is to demand first-person evidence nearby, but that also discards some correct detections

whose first-person marker is elsewhere in the discourse. Step 4 then showed a different kind of problem: a genuine *recall* bug, in which the dedup logic keyed on the noun type rather than on each occurrence, so that a second *kitaab* in the same segment could never be flagged.

### 8.4 Reconciling the "23 Manually Counted" Errors

An earlier, informally compiled catalogue in the repository listed 23 gender errors and had been quoted as the manual ground truth. Checking each entry against the verified transcripts showed that it cannot serve as ground truth in its current form. Five entries could not be located in any verified transcript segment (for example S8 *khursi toot gaya*, *kitaab ulat gaya*, *gaadi kharab ho gaya*); four are not gender-agreement errors under the audit criteria (*janam hua*; *hum kaam kiya* and *apne course liya*, where *kiya/liya* correctly agree with masculine *kaam/course*; and *mai study kiya*, which is contested); two have a mixed-gender group as subject; and one is a double count of the same segment. Only the remaining entries correspond to genuine detections. We therefore treat the 23-item list as superseded, and use the audited list in Appendix A as the reference.

### 8.5 Resulting Precision and Recall

After the fourth iteration the analyzer reports 21 gender errors, of which all 21 were judged correct on reading, i.e. precision 21/21 on the audited set; one of these (S4 *mera bhai beti english mein padh raha*) is borderline because *bhai beti* refers to a mixed group of "brothers and daughters", so a conservative reading gives 20/21 (95%). Two further genuine errors in the audited material are still missed: S4's *kaam karta tha* and *ghus gaya* in a narrative segment ("*Bhanwar hospital mein kaam karta tha toh idhar ghus gaya*") have no first-person marker anywhere near them, and attributing the verbs to the speaker requires discourse-level reasoning about who the narrative is about. This gives recall of 21/23 (91%) on the audited material. These figures should be read with two cautions: the rules were tuned on the same audit, so they are optimistic, and recall is measured only against errors found by the same reader who built the rules. A held-out set of speakers, annotated by a second reader, is needed for an unbiased estimate.

## 9 Evaluation

### 9.1 Analyzer Output on Corrected Transcripts

Running the final analyzer on the human-corrected transcripts:

Table 9: Analyzer output on corrected transcripts

| **Metric** | **Count** |
|---|---|
| Total words analyzed | 2,060 |
| *Language identification* | |
| Hindi | 67.1% |
| English | 12.7% |
| Unknown | 12.5% |
| Bengali (candidates) | 7.7% |
| Gender errors detected | 21 |
| Noun-verb (R1) | 12 |
| Subject-verb (R2) | 9 |
| Code-switches detected | 78 |
| Transitive perfective contexts | 8 |
| With *ne* | 1 |

The "unknown" language category (12.5%) consists primarily of proper nouns (Kharagpur, IIT), numbers, hesitation markers (*uhhh*), and bracket annotations (`[unclear]`). The lexicon covers 87.5% of tokens.

### 9.2 Language Distribution by Speaker

Table 10: Word-level language identification by speaker (% of tokens)

| **Spk.** | **Tokens** | **Hindi** | **English** | **Bengali** | **Unknown** |
|---|---|---|---|---|---|
| S1 | 178 | 60.7 | 14.6 | 8.4 | 16.3 |
| S2 | 202 | 52.5 | 10.4 | 16.8 | 20.3 |
| S3 | 230 | 71.7 | 17.8 | 0.9 | 9.6 |
| S4 | 307 | 76.5 | 11.1 | 2.9 | 9.4 |
| S5 | 307 | 67.8 | 12.1 | 7.5 | 12.7 |
| S6 | 299 | 61.2 | 8.7 | 17.7 | 12.4 |
| S7 | 308 | 69.5 | 14.9 | 5.2 | 10.4 |
| S8 | 229 | 71.2 | 13.5 | 3.1 | 12.2 |
| All | 2,060 | 67.1 | 12.7 | 7.7 | 12.5 |

The two highest-Bengali speakers are S6 (17.7%) and S2 (16.8%); the two lowest are S3 (0.9%) and S4 (2.9%). The ordering is not the same as the ordering by gender errors, which is why the profile in Section 5 treats Bengali use and Hindi accuracy as distinct dimensions: S4 uses little Bengali yet makes the most errors, while S6 uses the most Bengali and makes none in the structures the analyzer searches for.

### 9.3 Gender Errors by Speaker and Rule

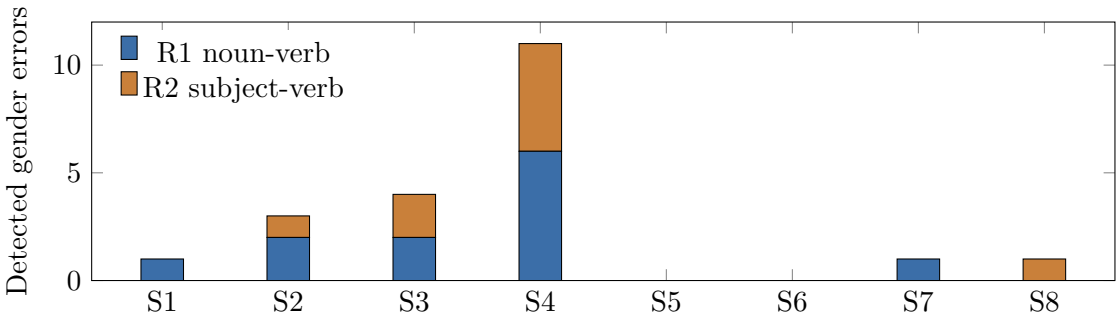



Figure 1: Verified gender-error detections per speaker, split by rule. R2 applies only to female speakers (S2, S3, S4, S8).

### 9.4 Code-Switch Typology by Speaker

Table 11: Code-switches by type and speaker (Poplack-style classification; Bengali runs are candidates, see Section 11.7)

| Spk. | Tag | Intra-sent. | Inter-sent. | Three-way | Total |
|---|---|---|---|---|---|
| S1 | 4 | 1 | 0 | 5 | 10 |
| S2 | 5 | 2 | 3 | 2 | 12 |
| S3 | 1 | 0 | 0 | 1 | 2 |
| S4 | 5 | 1 | 0 | 2 | 8 |
| S5 | 4 | 3 | 0 | 7 | 14 |
| S6 | 1 | 4 | 5 | 1 | 11 |
| S7 | 6 | 1 | 0 | 7 | 14 |
| S8 | 4 | 0 | 0 | 3 | 7 |
| **Total** | **30** | **12** | **8** | **28** | **78** |

Three patterns stand out. First, tag and three-way switches together account for 58 of 78 switches (74%): most Bengali material in the corpus is inserted at word level, not as full clauses. Second, inter-sentential switching is confined to two speakers, S6 (5) and S2 (3), who are also the two highest-Bengali speakers. Third, the *three-way* category—a Bengali element in the neighbourhood of English words—is the largest single category for S1, S5 and S7, reflecting that these speakers' Hindi frames are themselves rich in English loanwords (*chair, door, photo, shop*).

### 9.5 Ergative *ne* Contexts

The analyzer detects 8 candidate transitive perfective contexts (Appendix B): two for S1 (*padha* in "*bcom hum 2 year padha tha*" and "*padha toh bcom tak*"), one for S2 (*bola*), three for S4 (*padha*, *seekha*, *bola*) and two for S6 (*khola*, *seekhi*). One carries a fused *ne* (S6, *maine hindi …seekhi*). The most informative contexts are the first-person ones: in *hum jab 7th class aur 8th class padha aur do class mein hindi seekha* (S4) the standard form would be *maine …padha …seekhi* with the verb agreeing with feminine *hindi*; S4 instead uses the bare subject *hum* with a masculine verb. In this example the absence of *ne* goes together with a default masculine verb, which suggests that the two transfer phenomena may be linked, although one example cannot establish it. The sample is too small for any quantitative claim, and some contexts are not clean examples (*khola* is a Bengali adjective).

### 9.6 Section-Level Error Distribution

Table 12 breaks down detected phenomena by questionnaire section. This reveals where each transfer pattern is most active.

Table 12: Phenomena by questionnaire section

| Sec. | Task | Segs | Words | Gender Err. | Switches | Bengali |
|---|---|---|---|---|---|---|
| A | Naming | 18 | 270 | 3 | 11 | 36 |
| B | Adjective | 5 | 74 | 4 | 4 | 4 |
| C | Action | 11 | 206 | 3 | 16 | 35 |
| E | Conversation | 44 | 851 | 6 | 29 | 37 |
| F | Background | 42 | 659 | 5 | 18 | 47 |
| | **Total** | **120** | **2,060** | **21** | **78** | **159** |

Key observations:

- Section A (naming) elicits the most Bengali lexical items per word, confirming that household vocabulary triggers L1 access (H3).
- Section C (action descriptions) has the highest code-switch density (16 switches in 206 words = 7.8%), because speakers describe photos using Bengali verb forms.
- Section B (the adjective/comparison task) has the highest density of gender errors (4 in 74 words, 54 per 1,000), but from a tiny base; all four come from S4's list of states.
- Gender errors in the two conversational sections (E and F) are 11 of 21, i.e. over half; they are mostly R2 errors, which arise in spontaneous self-narrative where speakers are least monitoring their output.
- Section D (intransitive states) is subsumed into sections A and C in practice, as speakers describe broken/open/fallen objects within the same photo-description task.

## 10 ASR Comparison: Corrected vs. Whisper vs. IndicWhisper

We compare four text sources: human-corrected transcripts (the reference), Whisper-large-v3 raw output, and two IndicWhisper variants—`vasista22/whisper-hindi-small` (244M parameters) and `vasista22/whisper-hindi-large-v2` (1.5B parameters), both Hindi-finetuned Whisper models. The IndicWhisper models were run on the same audio, in 30-second chunks, on a laptop GPU through the `transformers` speech-recognition pipeline. All four texts are passed through the *same* analyzer so that differences in output reflect differences in the text, not the analysis. One caveat applies throughout: the ASR outputs do not separate collector from participant turns, so their word counts include the collector's speech and are measures of coverage rather than of accuracy; the comparison of *preserved phenomena* is unaffected, since the collector's speech is standard Hindi.

### 10.1 Main Results

Table 13: Four-way ASR comparison: impact on detected phenomena

| Phenomenon | Corrected | Whisper-v3 | Indic-small | Indic-large |
|---|---|---|---|---|
| Total words | 2,060 | 2,957 | 742 | 2,209 |
| Gender errors | 21 | 0 | 0 | 2 |
| Code-switches | 78 | 48 | 8 | 40 |
| Bengali words | 159 | 52 | 9 | 45 |
| Hallucination segs. | — | — | 2 | 6 |
| *Percentage of phenomena hidden (vs. corrected):* | | | | |
| Gender errors | — | 100% | 100% | 90% |
| Code-switches | — | 38% | 90% | 49% |
| Bengali words | — | 67% | 94% | 72% |

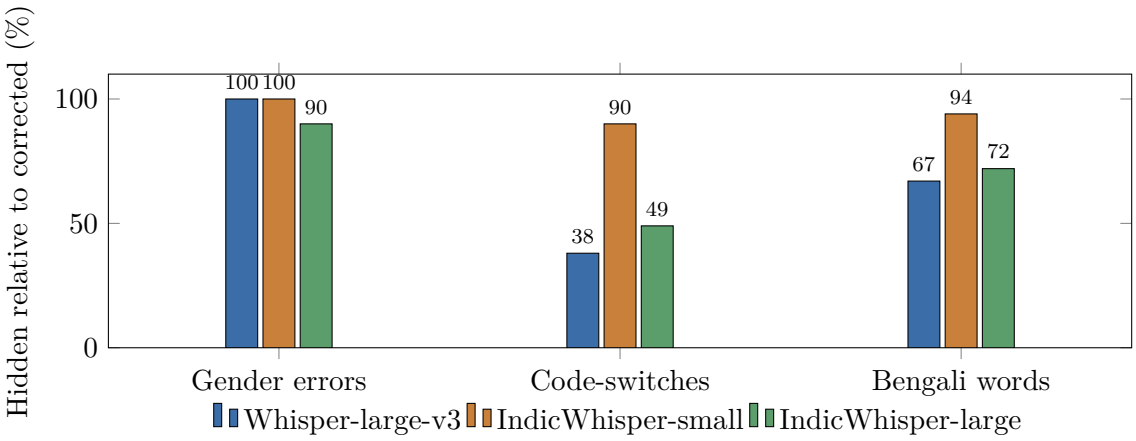



Figure 2: Share of each phenomenon hidden by each ASR system. Gender-error normalisation is near-total for all three; code-switch preservation is the only dimension on which the systems differ substantially.

**Key findings:**

1. **Gender error normalization is near-total across all models.** Whisper-large-v3 and IndicWhisper-small hide 100% of gender errors; IndicWhisper-large preserves only 2 of 21 (90% hidden). The two surviving errors are in S4 and S8, suggesting that the large model occasionally reproduces non-standard forms when the speaker's actual pronunciation is sufficiently clear, but this is unreliable.
2. **Model size matters for coverage but not for morphological fidelity.** IndicWhisper-large (2,209 words) achieves comparable coverage to Whisper-large-v3 (2,957 words), while IndicWhisper-small produces only 742 words—a 64% loss. However, even with adequate word counts, the large Hindi model still hides 90% of gender errors.
3. **Bengali word recognition improves with scale but remains poor.** IndicWhisper-large detects 45 Bengali words (72% loss) vs. IndicWhisper-small's 9 (94% loss). Whisper-large-v3, being multilingual, detects 52 (67% loss). Hindi-finetuned models convert Bengali words to Hindi-sounding equivalents more aggressively than the multilingual baseline.
4. **Hallucination increases with Hindi finetuning.** IndicWhisper-large produces 6 hallucination segments (repetition loops like "हाउस" ×60 and fabricated news-article text), while the small model produces 2 and Whisper-large-v3 produces none. Code-switched audio appears

to trigger the hallucination failure mode in Hindi-finetuned models, which expect monolingual input.

5. **Whisper-large-v3 remains the best ASR baseline overall**, with the highest code-switch preservation (48/78) and Bengali word detection (52/159), and no hallucinations. However, its 100% gender error normalization means it still cannot serve as a transparent frontend for morphological analysis.

These results demonstrate that the ASR normalization problem is **architectural, not a matter of model size or language-specific finetuning**. All three ASR configurations—multilingual large, Hindi-finetuned small, and Hindi-finetuned large—hide the majority of non-standard morphological forms. Human-corrected transcripts are necessary, not merely preferable, for studying L1 transfer in code-switched speech.

## 10.2 Per-Speaker Results

Table 14: Words and Bengali-word candidates by speaker and text source (C = corrected, W = Whisper-v3, S = Indic-small, L = Indic-large)

| | **Words** | | | | **Bengali words** | | | | **GE** |
|---|---|---|---|---|---|---|---|---|---|
| **Spk.** | C | W | S | L | C | W | S | L | C / L |
| S1 | 178 | 282 | 48 | 159 | 15 | 3 | 1 | 1 | 1 / 0 |
| S2 | 202 | 423 | 158 | 284 | 34 | 9 | 0 | 3 | 3 / 0 |
| S3 | 230 | 398 | 52 | 408 | 2 | 7 | 0 | 4 | 4 / 0 |
| S4 | 307 | 302 | 170 | 421 | 9 | 5 | 1 | 5 | 11 / 1 |
| S5 | 307 | 469 | 158 | 321 | 23 | 15 | 5 | 17 | 0 / 0 |
| S6 | 299 | 356 | 45 | 132 | 53 | 6 | 0 | 3 | 0 / 0 |
| S7 | 308 | 451 | 65 | 241 | 16 | 6 | 2 | 8 | 1 / 0 |
| S8 | 229 | 276 | 46 | 243 | 7 | 1 | 0 | 4 | 1 / 1 |
| **All** | 2,060 | 2,957 | 742 | 2,209 | 159 | 52 | 9 | 45 | 21 / 2 |

Three things are visible in Table 14. The small model's coverage collapses on the longest and most Bengali recordings (S1: 48 of 178 words; S6: 45 of 299). S6, the highest-Bengali speaker, loses most under every system (53 corrected Bengali words against 6, 0 and 3), which is what one would expect if the models are trained to produce Hindi. And S3, who uses almost no Bengali, shows *more* "Bengali" words in some ASR outputs than in the corrected text (7 and 4 against 2): these are Hindi words that the suffix heuristic mis-labels in noisier text, which illustrates the interaction between ASR output and a heuristic language-identifier that is discussed in Section 11.7.

## 10.3 Whisper Segment-Level Error Analysis

Measured at segment level against the human corrections (Section 4), Whisper-large-v3 needs correction in 120 of 144 participant-speech segments (83.3%). The error rate ranges from 60% (S2) to 100% (S1, S7, S8); the noisiest recording (S4) and the most Bengali recording (S6) are both at 94%.

Table 15: Whisper-large-v3 segment-level error rate by recording

| Recording | Segments | Corrected | Verified OK | Error rate | Bengali segs. missed |
|---|---|---|---|---|---|
| S1 (6647) | 11 | 9 | 0 | 100% | 1 |
| S1 (6648) | 9 | 5 | 0 | 100% | 1 |
| S2 | 30 | 12 | 8 | 60% | 6 |
| S3 | 27 | 15 | 5 | 75% | 1 |
| S4 | 18 | 15 | 1 | 94% | 3 |
| S5 | 34 | 17 | 9 | 65% | 5 |
| S6 | 21 | 15 | 1 | 94% | 12 |
| S7 | 20 | 19 | 0 | 100% | 5 |
| S8 | 13 | 13 | 0 | 100% | 3 |
| **Total** | **183** | **120** | **24** | **83.3%** | **37** |

(Segment counts include collector-only and noise segments; the error rate is corrected divided by corrected plus verified-OK.) Of 28 distinct Bengali words appearing in the corpus (71 instances), Whisper detected none as Bengali: all were converted to Hindi equivalents or to garbled Devanagari.

### 10.4 How the Models Rewrite Bengali

Table 16: Examples of Bengali material rewritten by Whisper-large-v3

| Spoken (Bengali) | Whisper output | Meaning |
|---|---|---|
| *dorja* | *darwaza* (Hindi equivalent) | door |
| *janala* | *khidki* (Hindi equivalent) | window |
| *jol* | *paani* (Hindi equivalent) | water |
| *bhat* | *chawal* (Hindi equivalent) | rice |
| *pakha* | *pankha* (Hindi equivalent) | fan |
| *cheleti porachuna korche* | garbled Devanagari | the girl is studying |
| *boi ulto kore rekh chi* | garbled Devanagari | kept the book upside down |

The pattern is that single lexical items are replaced by the Hindi equivalent (so the switch is deleted but the sentence stays grammatical) while whole Bengali clauses, which have no Hindi equivalent form, are rendered as nonsense. Both outcomes remove the evidence of code-switching; the first does so silently.

### 10.5 Hallucination Loops

Table 17: Repetition loops detected in IndicWhisper output (repeated $n$-gram $\geq 5$ times)

| Speaker | Model | Seg. | Repeated phrase | Repeats |
|---|---|---|---|---|
| S2 | small | 11 | दो हजार | 30 |
| S4 | small | 9 | थोड़ा थोड़ा | 67 |
| S2 | large | 1 | हाउस हाउस | 60 |
| S2 | large | 3 | रहा है | 9 |
| S3 | large | 9 | बढिया बढिया | 52 |
| S3 | large | 12 | थोड़ा थोड़ा | 52 |
| S4 | large | 3 | रहा है | 7 |
| S4 | large | 9 | थोड़ा थोड़ा | 53 |

Hallucination in the large model is not limited to loops. In S6's recording, the large model replaces a stretch of Hindi-Bengali speech with fluent but fabricated news-agency prose about a politician's statement, which cannot be detected by repetition and was found only by reading. For a downstream analysis this is arguably the worst case: the output is well-formed Hindi that bears no relation to the audio.

## 11 Discussion

### 11.1 Evaluating the Hypotheses

**H1 (Transfer): supported, with a caveat.** All 21 verified gender errors are masculine-default. Because the analyzer searches only for masculine forms where feminine is expected, this on its own is partly circular. We therefore also scanned all participant speech for feminine verb forms. Most are correct: *khud banati hu*, *rehti*, *padhati hu*, *karti hu*, *aa jati hu*, *student use karti hai*, *maine hindi seekhi*. One is a candidate reverse error, S3's *mera bada ladka school mein thi* (feminine *thi* for masculine *ladka*), which could be an agreement slip, a mis-hearing for *ladki*, or an overgeneralisation. The honest summary is therefore 21 masculine-default errors against one candidate feminine-on-masculine instance, rather than the absolute unidirectionality sometimes claimed.

**H2 (Avoidance): partially supported.** Of the eight candidate ergative contexts only one contains *ne* (S6, *maine*). Seven of eight is consistent with avoidance, but the denominator is tiny and one context is not a Hindi perfective, so the result should be read as suggestive.

**H3 (Salience): supported.** Bengali is concentrated in the naming section (36 Bengali words in 270 words, 13.3% of tokens, against 4.3% in the conversation section and 7.1% in the background section) and is dominated by household nouns.

**H4 (Opacity): supported.** Section 10 shows that three ASR systems hide 90–100% of gender errors. The hypothesis is about the output of the systems; it does not claim that they could not be modified (for example by finetuning on code-switched data), only that the available Hindi-finetuned models do not.

### 11.2 Variable Rather than Categorical Masculine Default

Several female speakers produce both correct and incorrect forms in the same recording: S3 says *mai khud banati hu* but *hum log karta hai*; S8 produces four correct feminine first-person habituals and one masculine (*mai hi banata hai*); S4's *mai* is more consistently masculine. This is the pattern one would expect from partial acquisition: speakers know the feminine forms, and the masculine form appears as a default under production pressure rather than as a complete absence of the

category. The error rate per speaker is therefore a measure of the *rate of defaulting*, not of whether the category is known, which is consistent with the interlanguage view of L2 development.

### 11.3 Proficiency Gradient

Error frequency correlates inversely with Hindi exposure:

- S4 (11 errors, 35.8 per 1,000 words): Hindi learned in school, Bengali the main language among colleagues, noisy recording.
- S3 (4 errors) and S2 (3 errors): intermediate; S3 uses almost no Bengali but still defaults on self-reference.
- S8 (1 error): mostly correct feminine self-reference; shortest recording.
- S5, S6 (0 errors): male speakers—masculine default is correct for self-reference. S5 reports years of residence in North India; S6 learned Hindi at the institute and produces the most Bengali.

The gradient should be read as a description of this sample, not as a causal claim: with eight speakers, other explanations (age, task effects, noise in S4's recording) cannot be separated.

### 11.4 Avoidance as Strategy

The near-total absence of ergative *ne* (1 of 8 candidate contexts) is consistent with an avoidance strategy, not ignorance. Speakers restructure sentences or use bare-subject forms (*hum …padha*) rather than producing incorrect *ne* constructions. This is linguistically more sophisticated than error production and consistent with the Avoidance Hypothesis [Schachter, 1974]. The only attested *ne* (S6) is accompanied by correct object agreement (*seekhi*), which suggests that when the construction is produced it is produced accurately.

### 11.5 Insertion or Alternation?

In Muysken's typology, the dominance of single-word and short-run Bengali material inside a Hindi or English frame (58 of 78 switches are tag or three-way) is characteristic of *insertion*, with Hindi as the matrix language. The inter-sentential switches of S6 and S2, in which whole Bengali clauses alternate with Hindi, are characteristic of *alternation*. The corpus thus contains both, and the balance depends on the speaker: it looks like a continuum from insertional (S3, S4, S8) to alternational (S6), with S2 and S5 in between. We note that this classification is based on lexicon-detected runs and should be confirmed by manual annotation.

### 11.6 Implications for ASR and NLP

- **Evaluation.** Word error rate against a standardised reference rewards exactly the normalisation that destroys morphological evidence. A system that turns *kitaab padh raha* into *kitaab padh rahi lowers* its WER against a standard reference while erasing the speaker's grammar. Evaluation of ASR for linguistically diverse speakers should include a measure of preservation of non-standard forms.
- **Data.** Hindi-finetuned models are trained on standard Hindi; our results indicate that fine-tuning on standard data entrenches normalisation rather than reducing it. Models intended to transcribe code-mixed speech faithfully need training data in which the mixing and the non-standard morphology appear as targets.
- **Pipelines.** Until such models exist, studies of L1 transfer must keep a human in the loop, and the human correction step should be treated as part of the instrument and its cost reported.

### 11.7 Limitations

1. **Corpus size**: 8 speakers and 2,060 words is small for statistical generalization, though sufficient for documenting qualitative patterns. Counts per speaker (1–11 errors) are too low for significance testing.
2. **Rule-based approach**: The analyzer cannot detect novel patterns not encoded in its lexicons and rules. A machine learning approach would require a larger annotated corpus.
3. **Unknown words**: 12.5% of words are classified as "unknown," primarily proper nouns and numbers outside the lexicon's scope.
4. **Gender error precision and recall**: The audit (Section 8) gives precision 21/21 (or 20/21 conservatively) and recall 21/23 on the audited material, but the rules were tuned on that same material by a single annotator. The figures are optimistic and should be validated on held-out speakers with a second annotator.
5. **Language-identification precision**: The Bengali-word and code-switch counts (159 words, 78 switches) are *candidate* counts. The suffix and marker heuristics also match some Hindi words: for example *ke* and *khana* (listed Bengali markers) and words ending in *-le/-che/-chi* such as *bole, pehle, role, acchi, bachche*, and the Hindi numeral *che* "six" in *che mahine*, were flagged as Bengali in some segments. The true counts of Bengali words and of code-switches are therefore lower than reported, and the counts should be treated as upper bounds. A manual audit of language identification, analogous to the gender audit, has not yet been done. The *relative* patterns (S6 and S2 highest, S3 lowest; naming section highest) are robust to this because the false-positive words are spread across speakers, but the absolute figures are not.
6. **Ergative contexts**: The eight candidate contexts have not been manually audited. At least one (*khola*) is not a Hindi perfective, so the denominator may be seven or fewer.
7. **Elicitation effects**: Many errors arise in the picture-description tasks, where the speaker describes a third party; R1 is therefore partly a function of the task. R2 errors arise in free conversation and are less task-dependent.
8. **Acoustic quality**: S4's recording is the noisiest, and misheard forms there would be indistinguishable from errors in the transcript; the high S4 error count should be read with this in mind.
9. **ASR comparison**: ASR word counts include the collector's speech (Section 10). The comparison runs all texts through the same lexicon-based analyzer. Part of the ASR systems' apparent loss of Bengali words is genuine, but part is due to differences in orthography after transliteration from Devanagari. The direction of the result (heavy normalisation) is large and consistent enough to be robust, but the exact percentages are not precise.

## 12 Conclusion and Future Work

We have presented a morphology-aware analyzer for Bengali-Hindi code-switched speech that detects L1 transfer patterns invisible to standard ASR, together with a documented audit of its precision. The study demonstrates that:

1. Gender agreement errors in L2 Hindi are **systematic and predictable** from the L1 (Bengali) typology: the 21 verified errors are all masculine-default, against a single candidate reverse instance, and they are variable rather than categorical within speakers.
2. **ASR normalization is architectural, not fixable by finetuning.** A four-way comparison (human-corrected, Whisper-large-v3, IndicWhisper-small, IndicWhisper-large) shows that the ASR systems hide 90–100% of gender errors. Hindi finetuning introduces hallucination on code-switched audio without preserving non-standard morphology.
3. A rule-based analyzer with curated lexicons can detect these patterns with high precision on

corrected transcripts, but its precision depends on lexical knowledge (noun gender, verb/noun status) and on discourse context (who the subject is), and must be measured by audit rather than assumed. The audit moved the analyzer from 53% initial precision to a verified list of 21 detections.

4. The same discipline needs to be applied to language identification and ergative detection, whose counts are currently candidates rather than verified numbers.

### 12.1 Future Directions

- **Held-out validation**: Collect recordings from additional speakers and have two independent annotators label them, so that precision, recall and inter-annotator agreement can be estimated without the tuning bias in the present audit.
- **Language-identification audit**: Manually verify the Bengali-word and code-switch candidates, replace the suffix heuristic with context-sensitive rules, and report corrected counts.
- **Hindi L1 comparison**: Add one or more Hindi L1 speakers from the same workplace as a comparison group, to confirm that the observed patterns are specific to Bengali L1 rather than to the elicitation tasks.
- **Larger corpus**: Expanding to more speakers and longer recordings for statistical analysis.
- **Other L1-Hindi pairs**: The framework is extensible to other genderless L1s (e.g., Tamil, Telugu) transferring into Hindi.
- **Neural approach**: Training a sequence labeling model on the annotated data for joint language ID and error detection, and finetuning ASR on the corrected transcripts to test whether non-standard forms can be preserved.
- **Discourse-aware R2**: Resolving who the subject of a narrative is, so that self-referential verbs without a nearby first-person marker (the two residual misses) can be detected.
- **Real-time feedback**: Integrating the analyzer into a language learning tool for Bengali speakers learning Hindi.

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# A Verified Gender-Error Detections

Table 18 lists all 21 detections of the final analyzer with the questionnaire section of the segment in which each occurs. "Segment" is the segment number within the speaker's transcript.

Table 18: All verified gender-error detections (final analyzer)

| # | Spk. | Seg. | Sec. | Rule | Produced | Expected |
|---|---|---|---|---|---|---|
| 1 | S1 | 2 | A | R1 | *roti kha raha* | *kha rahi* |
| 2 | S2 | 4 | C | R1 | *roti kha raha* | *kha rahi* |
| 3 | S2 | 4 | C | R1 | *kitaab padh raha* | *padh rahi* |
| 4 | S2 | 8 | E | R2 | *seekh raha hu* | *seekh rahi hu* |
| 5 | S3 | 6 | A | R1 | *kitaab padh raha* | *padh rahi* |
| 6 | S3 | 7 | A | R1 | *chair tut gaya* | *tut gayi* |
| 7 | S3 | 10 | E | R2 | *hum log karta hai* | *karti hai* |
| 8 | S3 | 15 | F | R2 | *bol nahi pata hu* | *pati hu* |
| 9 | S4 | 1 | B | R1 | *almari khul raha* | *khul rahi* |
| 10 | S4 | 1 | B | R1 | *kitaab padh raha* | *padh rahi* |
| 11 | S4 | 1 | B | R1 | *kitaab ulta ho gaya* | *ulti ho gayi* |
| 12 | S4 | 1 | B | R1 | *chair tut gaya* | *tut gayi* |
| 13 | S4 | 3 | E | R1 | *kamar tut gaya* | *tut gayi* |
| 14 | S4 | 3 | E | R2 | *seekh raha* | *seekh rahi* |
| 15 | S4 | 3 | E | R2 | *ghus gaya* | *ghus gayi* |
| 16 | S4 | 3 | E | R2 | *a gaya* | *a gayi* |
| 17 | S4 | 5 | F | R1 | *beti padh raha*[†] | *padh rahi* |
| 18 | S4 | 16 | F | R2 | *bengali mein hi bolta hai* | *bolti hai* |
| 19 | S4 | 17 | F | R2 | *baat karta hai* | *karti hai* |
| 20 | S7 | 4 | C | R1 | *ladies door khol raha* | *khol rahi* |
| 21 | S8 | 8 | F | R2 | *mai hi banata hai* | *banati hai* |

[†]Borderline: subject is the mixed group *bhai beti*.

# B Ergative Candidate Contexts

Table 19: Candidate transitive perfective contexts (R3)

| Spk. | Seg. | Verb | *ne*? | Context |
|---|---|---|---|---|
| S1 | 9 | *padha* | no | *bcom hum 2 year padha tha* |
| S1 | 10 | *padha* | no | *padha toh bcom tak hi english* |
| S2 | 8 | *bola* | no | *…usko seekh ke sunne se ha woh yeh bola* |
| S4 | 10 | *padha* | no | *hum jab 7th class aur 8th class padha* |
| S4 | 10 | *seekha* | no | *…do class mein hindi seekha* |
| S4 | 15 | *bola* | no | *…woh ghusa diya bola idhar hi kaam karo* |
| S6 | 11 | *khola* | no | *janala khola [unclear] ache* (Bengali adjective "open") |
| S6 | 13 | *seekhi* | **yes** | *maine hindi iit mein rehke seekhi* |

# C Audit of the 36 Initial Detections

Of the 36 initial detections, the following 17 were judged false positives. (Numbers are those of the original audit listing, in order of speaker and segment.)

Table 20: The 17 false positives of the initial analyzer

| # | Spk. | Phrase | Cause | Reason |
|---|---|---|---|---|
| 3 | S2 | *chawal kha raha hai* | lexicon | *chawal* is masculine |
| 6 | S3 | *paani daal raha hai* | homograph | *daal* is the verb "pour" |
| 7 | S3 | *glass …raha hai* | lexicon | *glass* is masculine |
| 9 | S3 | *janam hua* | impersonal | birth happened; *janam* masculine |
| 12 | S4 | *glass tut gaya* | lexicon | *glass* is masculine |
| 13 | S4 | *pitaji gujar gaya* | third person | subject is the speaker's father |
| 19 | S4 | *seekh raha* (repeat) | duplicate | same verb already counted |
| 20 | S4 | *woh padhta hai* | third person | *woh* is a male child |
| 22 | S4 | *padh raha* (R2) | duplicate | already counted as R1 |
| 23 | S4 | *padhta hai* | third person | about someone's child |
| 24 | S4 | *woh kya bolta hai* | third person | *woh* is another person |
| 27 | S4 | *karta tha* (repeat) | duplicate | same verb already counted |
| 30 | S4 | *kaam karta tha* | mixed group | *lerka lerki* (boys and girls) |
| 31 | S4 | *padhta tha* | mixed group | *lerka lerki* |
| 32 | S4 | *karta tha* | mixed group | *lerka lerki* |
| 33 | S4 | *padhta* | mixed group | *lerka lerki* |
| 36 | S8 | *non-veg hota hai* | impersonal | generic statement about food |

The other 19 detections were true positives. After the four iterations of Table 8, they reappear in Appendix A together with the recall repairs.

# D Reproducing the Results

The repository is organised as follows: `transcripts/` holds the per-recording transcript JSON (Whisper text and corrected text per segment) and the IndicWhisper outputs; `verification/` holds the human correction records; `model/` holds the analyzer, the lexicons, the transliterator and the section labeller; `analysis/` holds all generated outputs; and `scripts/` holds the extraction, transcription, comparison and IndicWhisper scripts. The commands below regenerate every table in Sections 9 and 10.

```
python3 run_analysis.py                    # per-speaker JSON, corpus summary,
```

```
                                        # evaluation report, section labels
python3 scripts/03_linguistic_analysis.py --use-corrections \
        transcripts analysis            # per-recording analyses
python3 scripts/compare_indicwhisper.py # four-way ASR comparison
python3 scripts/run_indicwhisper.py     # (optional) rerun IndicWhisper
                                        # on a machine with a GPU
```

All steps are deterministic. This report is compiled with XeLaTeX (the Devanagari examples require a Unicode engine).